

浙江大学

ZJUI 学院 2025 年暑期科研项目总结材料



中文论文题目：基于平均流模型的一步去噪

英文论文题目：One-step Denoising with MeanFlow Model

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摘要

本工作探索了平均流一步生成式模型 (MeanFlow) 在图像去噪任务中的应用。本研究的主要目标是理解流匹配生成式模型 (Flow Matching) 及其优化方案 MeanFlow 模型的基本原理,并在 MNIST 与 CIFAR-10 数据集上进行实验。在此基础上,我们将 MeanFlow 模型改造用于图像去噪任务:不再以简单的高斯分布作为初始分布,而是以加噪声的真实图像作为起点,通过 MeanFlow 模型的一步生成过程对图像进行去噪。MNIST 实验表明,一次函数评估 ($\text{NFE} = 1$) 即可显著降低噪声,使峰值信噪比 (PSNR) 从约 6 dB 提升至 22.6 dB。二次函数评估 ($\text{NFE} = 2$) 将 PSNR 进一步提升至约 24 dB,而更多推理步骤增益有限。实验结果表明,MeanFlow 模型能以极低的推理成本有效完成图像去噪任务。本工作未提出新的算法或模型框架,而是对一步生成模型在图像去噪任务中的应用进行了实践探索,初步验证了 MeanFlow 方法在快速图像去噪中的可行性。

Abstract

This work investigates the application of the MeanFlow one-step generative model to image denoising tasks. The primary objective of this summer research is to understand the fundamental principles of Flow Matching generative modeling and its improved version, the MeanFlow model, through theoretical derivation and experiments. Based on this understanding, we specifically adapt the MeanFlow model for image denoising by modifying its initial distribution: from simple distributions like Gaussian to noisy images, retaining the model’s one-step generation capacity while serving the denoising purpose. Experiments on MNIST dataset show that a single function evaluation ($\text{NFE} = 1$) significantly reduces the noise, increasing the Peak Signal-to-Noise Ratio (PSNR) from approximately 6 dB to 22.6 dB. A second evaluation ($\text{NFE} = 2$) further improves the PSNR to 24 dB, while additional sampling steps yield limited gains. This work demonstrates that the MeanFlow framework is capable of performing effective image denoising at minimal sampling cost, positioning it as a practical alternative to conventional multi-step denoising methods. Although our work does not propose new algorithms or model frameworks, it explores the application of the MeanFlow framework to image denoising, verifying its feasibility for future image restoration tasks.

正文编排格式

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